

COURSE DESCRIPTION			
Program	Architecture, Automobile Engineering, Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Fire Technology and Safety, Interior Design, Mechanical Engineering (Automobile Engg.), Mechatronics, Mechanical Engineering, Civil Engineering (Construction Technology), Tool & Die Engineering, Wood and Paper Technology		
Course Title	Mathematics for Industrial Engineering		
Course Code	2001A	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	3:1:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces fundamental mathematical tools essential for engineering studies. It includes complex numbers, probability, statistics, integral calculus, differential equations, vector algebra and some techniques in numerical analysis. These topics help students to understand and solve engineering problems.

Course Objectives

1	To provide comprehensive introductory-level coverage of complex numbers
2	To develop an understanding of Probability and Statistics.
3	To introduce the fundamentals of integral calculus and differential equations for modelling and solving engineering-related problems.
4	To familiarize students with the basics of numerical methods and vector algebra required for problem solving.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Trigonometry, Determinants, Differentiation	1002	Fundamentals of Engineering Mathematics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Apply the concepts related to Probability and Statistics in various mathematical problems.	Apply
CO3	Use integration techniques and differential equations to solve basic engineering problems.	Apply
CO4	Apply basic numerical methods and vector algebra techniques to solve simple engineering problems.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	-	-	-	-	-	-	-	-	-	-
Correlation Level	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	1	0	0	5.5	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Complex Numbers	The Complex Number System, Fundamental operations with complex numbers, Modulus and Amplitude, Graphical representation of complex numbers, Polar form	11	4
Module II	Probability and Statistics	Probability: Sets, Random experiments, Outcomes, Sample space, Events, Classical definition of probability, Axioms of probability. Statistics: Introduction to Statistics, Grouped and ungrouped data, Frequency distribution, Measures of Central Tendency.	12	3
Module III	Integral Calculus	Integration as the reverse process of differentiation, Standard results, Integration by parts, Integration by substitution integral of $f(ax+b)$, $f(x) * f'(x)$ and $f'(x)/f(x)$, Definite integrals, Area bounded by a curve and the x-axis.	11	4
Module IV	Numerical Methods and Vector Algebra	Numerical Methods: Interpolation, Lagrange's Method, Numerical Integration- Simpsons 1/3rd Rule Vector Algebra: Scalars and vectors, vector addition, subtraction, multiplication by a scalar, Dot product, Cross product	11	4

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
The complex number system	Understand the complex number system and identify the real part and imaginary part	L	CO1	Understand	1
Conjugate of a complex number, Equality of complex numbers	Determine the conjugate of a complex number, Test the equality of complex numbers by comparing their real and imaginary parts.	L	CO1	Apply	1
Fundamental operations with complex numbers : Addition, Subtraction, Multiplication, Division	Perform fundamental operations on complex numbers	L	CO1	Understand	5
Graphical Representation	Represent complex numbers graphically on the Argand (complex) plane.	L	CO1	Understand	1
Absolute value and Amplitude of a Complex Number	Determine the Modulus (Absolute value) and amplitude (argument) of a complex number in degrees	L	CO1	Apply	2
Polar Form	Express a complex number in polar form	L	CO1	Understand	1
Problems	Solve simple engineering problems involving complex numbers.	T	CO1	Apply	4
Module II					
Sets, Types of sets: Empty set and Universal set, subset of a set.	Define sets and identify empty sets, universal sets, and subsets.	L	CO2	Understand	1
Random Experiments, Outcomes, Sample space, events, classical definition of probability, axioms of probability	Explain random experiments, outcomes, sample spaces, events, and the axioms of probability.	L	CO2	Understand	3
simple numerical problems based on Tossing of coins (1 coin and 2 coins), throwing of a die and drawing of a card.	Construct sample spaces and solve simple probability problems.	L	CO2	Apply	2
Introduction to Statistics, Ungrouped data, Frequency distribution	Define statistics, organize ungrouped data, and construct frequency distributions.	L	CO2	Understand	2
Measures of Central Tendency: mean, median and mode. Simple problems based on ungrouped data.	Explain mean, median, and mode and compute them for ungrouped data.	L	CO2	Apply	2

Finding the mean, median and mode from an ungrouped frequency distribution	Calculate mean, median, and mode from an ungrouped frequency distribution.	L	CO2	Apply	2
Problems	Solve problems related to probability and statistics	T	CO2	Apply	3
Module III					
Integration as the reverse process of differentiation	Understand integration as the inverse process of differentiation	L	CO3	Remember	1
Integral of standard functions(x^n , e^x , $1/x$, $\sin x$, $\cos x$, $\sec^2 x$, $\operatorname{cosec}^2 x$, $\sec x \tan x$, $\operatorname{cosec} x \cot x$)	Remember standard integrals of basic functions	L	CO3	Remember	1
Rules of integration and simple problems, Integration by substitution of $f(ax+b)$, $f(x) \cdot f'(x)$ and $f'(x)/f(x)$	Use basic rules of integration (constant multiple, sum and difference rules) to solve simple integrals.	L	CO3	Understand	2
Integration by parts rule and problems	Apply the integration by parts method to evaluate integrals of product of two functions	L	CO3	Apply	2
Definite integrals and simple problems	Evaluate definite integrals	L	CO3	Apply	2
Differential equations, order and degree of D.E	Identify and classify differential equations by determining their order and degree.	L	CO3	Understand	1
First order differential equations in variable separable form, First order L.D.E, $dy/dx + Py = Q$	Solve first order differential equations (variable separable and linear)	L	CO3	Apply	2
Problems	Evaluate integrals and solve first order differential equations	T	CO3	Apply	4
Module IV					
Introduction, Interpolation (Definition),	Explain the concepts of interpolation	L	CO4	Understand	1
Lagrange's Method (Simple Problems, Avoid to find the polynomials)	Apply Lagrange's method to find missing values from given data.	L	CO4	Apply	1
Numerical Integration, Simpsons 1/3rd Rule (Simple Problems)	Apply Simpson's 1/3rd rule to evaluate simple definite integrals.	L	CO4	Apply	1
Concept of scalars and vectors, Types of vectors, The unit vectors i, j, k , Position vector of a point, Representation of a vector in terms of i, j, k	Distinguish between scalar and vector quantities and identify components of a vector	L	CO4	Understand	1
Magnitude of a vector	Determine magnitude of a vector and find the unit vector in the direction of a given vector	L	CO4	Understand	1
Addition and subtraction of vectors, Multiplication by a scalar	perform addition and subtraction of vectors, multiplication of vectors by a scalar	L	CO4	Understand	2
Dot Product	Compute the dot product of vectors and apply it to determine angle between vectors and perpendicularity	L	CO4	Apply	2
Cross Product	evaluate the cross product of vectors, parallelism	L	CO4	Apply	2
Problems	solve problems on numerical methods and vector algebra	T	CO4	Apply	4

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
	Find conjugate of complex numbers, Determine modulus and amplitude and write polar form of complex numbers	4
2	Solve problems related to equality, addition and subtraction of complex numbers	4
3	Multiply and divide complex numbers	4

4	Find the applications of complex numbers in respective engineering areas.	3
Module II		
5	Problems of finding probabilities of coins, die and cards	4
6	Problems of finding mean, median and mode	4
7	Problems of finding mean, median and mode from a frequency table	4
8	Find the applications of statistics and probability in respective engineering areas.	3
Module III		
9	Find integrals of functions	4
10	Evaluate definite integrals	4
11	Solve first order differential equations	4
12	Find the applications of integral calculus in respective engineering areas.	3
Module I		
13	Problems of interpolation and Simpson's $\frac{1}{3}$ rd rule Identify scalars and vectors,	5
14	Solve problems on vector addition and subtraction, Multiply vectors by a scalar, Perform dot product and cross product of vectors	5
15	Solve problems to find area and work done	5
Total Self-Learning hours		60

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Mathematics for Industrial Engineering		SITTTR Kalamassery

REFERENCE			
Sl. No.	Title	Author	Publication
1	Higher Engineering Mathematics	B. S. Grewal	Khanna Publishers, New Delhi
2	Higher Engineering Mathematics	B. V. Ramana	Tata McGraw-Hill Education, New Delhi
3	A text book of Engineering Mathematics	N. P. Bali & Manish Goyal	Laxmi Publications (P) Ltd., New Delhi
4	Schaum's outlines of Discrete Mathematics(3rd Edition)	Seymour Lipschutz & Marc Lipson	Mc Graw Hill
5	Schaum's outlines of Probability(3rd Edition)	Seymour Lipschutz & Marc Lipson	Mc Graw Hill
6	Schaum's outlines of Statistics(6th Edition)	Dr. Murray R. Spiegel & Larry J. Stephens	Mc Graw Hill

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	

Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

SITTTR